Impact Influence

Configuration des équipements

1. Plan d'adressage IP :

Groupes/Lien	VLAN ID	Adresse Réseau (IPV4)	Adresse Réseau (IPV6)	Passerelle Réseau	DHCP/STATIQUE
Direction	10	192.168.10.0/24	2001:0:CAFE:10::/64	192.168.10.254	DHCP
RH	20	192.168.20.0/24	2001:0:CAFE:20::/64	192.168.20.254	DHCP
Commercial	30	192.168.30.0/24	2001:0:CAFE:30::/64	192.168.30.254	DHCP
Finance	40	192.168.40.0/24	2001:0:CAFE:40::/64	192.168.40.254	DHCP
Informatique	50	192.168.50.0/24	2001:0:CAFE:50::/64	192.168.50.254	DHCP
Accueil	60	192.168.60.0/24	2001:0:CAFE:60::/64	192.168.60.254	STATIQUE
Showroom	70	192.168.70.0/24	2001:0:CAFE:70::/64	192.168.70.254	STATIQUE
Réunion	80	192.168.80.0/24	2001:0:CAFE:80::/64	192.168.80.254	STATIQUE
Serveur DNS	110	192.168.110.0/24	2001:0:CAFE:110::/64	192.168.110.254	STATIQUE
Serveur Web	120	192.168.120.0/24	2001:0:CAFE:120::/64	192.168.120.254	STATIQUE
Lien Switch_Distributed - Switch Core	150	192.168.150.0/30	2001:0:CAFE:150::/64		
Lien Switch_Distributed(2) - Switch Core	160	192.168.160.0/30	2001:0:CAFE:160::/64		
Switch Core - Routeur ROUGE	170	192.168.170.0/30	2001:0:CAFE:170::/64		
Lien Wan Routeur BLEU <-> Routeur ROUGE	N/A	132.186.32.32/30	2001:0:CAFE:200::/64		
Lien Wan Routeur VERT <-> Routeur ROUGE	N/A	132.186.32.36/30	2001:0:CAFE:210::/64		
Lien Wan Routeur BLEU <-> Routeur VERT	N/A	132.186.32.40/30	2001:0:CAFE:220::/64		

2. Configuration DNS :

No.	Name	Type	Detail
0	impactinfluence.com	A Record	132.186.32.42
1	impactinfluence6.com	AAAA Record	2001:0:CAFE:120::1

3. Configuration des routeurs (comprenant les tables de routage, les sous-interfaces et les NAT) :

Bâtiment Bleu :

Routage :

```
Routeur-BLEU#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       132.186.32.32/30 is directly connected, Serial0/3/1
L       132.186.32.33/32 is directly connected, Serial0/3/1
O       132.186.32.36/30 [110/128] via 132.186.32.34, 00:48:42, Serial0/3/1
                         [110/128] via 132.186.32.42, 00:48:42, Serial0/3/0
C       132.186.32.40/30 is directly connected, Serial0/3/0
L       132.186.32.41/32 is directly connected, Serial0/3/0
     192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/24 is directly connected, GigabitEthernet0/0.60
L       192.168.60.254/32 is directly connected, GigabitEthernet0/0.60
     192.168.70.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.70.0/24 is directly connected, GigabitEthernet0/1.70
L       192.168.70.254/32 is directly connected, GigabitEthernet0/1.70
```

```
Routeur-BLEU#show ipv6 route
IPv6 Routing Table - 20 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
O   2001:0:CAFE:10::/64 [110/67]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
O   2001:0:CAFE:20::/64 [110/67]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
O   2001:0:CAFE:30::/64 [110/67]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
O   2001:0:CAFE:40::/64 [110/67]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
O   2001:0:CAFE:50::/64 [110/67]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
C   2001:0:CAFE:60::/64 [0/0]
     via GigabitEthernet0/0.60, directly connected
L   2001:0:CAFE:60::254/128 [0/0]
     via GigabitEthernet0/0.60, receive
C   2001:0:CAFE:70::/64 [0/0]
     via GigabitEthernet0/1.70, directly connected
L   2001:0:CAFE:70::254/128 [0/0]
     via GigabitEthernet0/1.70, receive
O   2001:0:CAFE:80::/64 [110/65]
     via FE80::20A:41FF:FE32:B0DE, Serial0/3/0
O   2001:0:CAFE:110::/64 [110/67]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
O   2001:0:CAFE:120::/64 [110/65]
     via FE80::20A:41FF:FE32:B0DE, Serial0/3/0
O   2001:0:CAFE:150::/64 [110/66]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
O   2001:0:CAFE:170::/64 [110/65]
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
C   2001:0:CAFE:200::/64 [0/0]
     via Serial0/3/1, directly connected
L   2001:0:CAFE:200::1/128 [0/0]
     via Serial0/3/1, receive
O   2001:0:CAFE:210::/64 [110/128]
     via FE80::20A:41FF:FE32:B0DE, Serial0/3/0
     via FE80::250:FFF:FEAE:588C, Serial0/3/1
C   2001:0:CAFE:220::/64 [0/0]
     via Serial0/3/0, directly connected
L   2001:0:CAFE:220::1/128 [0/0]
     via Serial0/3/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

Sous-interfaces :

```
interface GigabitEthernet0/0.60
 description Vlan 60 - ACCUEIL
 encapsulation dot1Q 60
 ip address 192.168.60.254 255.255.255.0
 ip nat inside
 ipv6 address 2001:0:CAFE:60::254/64
 ipv6 ospf 1 area 0
!

 interface GigabitEthernet0/1.70
  description Vlan 70 - SHOWROOM
  encapsulation dot1Q 70
  ip address 192.168.70.254 255.255.255.0
  ip nat inside
  ipv6 address 2001:0:CAFE:70::254/64
  ipv6 ospf 1 area 0
```

```
interface Serial0/3/0
 description Lien Routeur-BLEU Routeur-VERT
 ip address 132.186.32.41 255.255.255.252
 ip nat outside
 ipv6 address 2001:0:CAFE:220::1/64
 ipv6 ospf 1 area 0
 clock rate 2000000
!
interface Serial0/3/1
 description Lien Routeur-BLEU Routeur-ROUGE
 ip address 132.186.32.33 255.255.255.252
 ip nat outside
 ipv6 address 2001:0:CAFE:200::1/64
 ipv6 ospf 1 area 0
 clock rate 2000000
```

NAT :

```
ip nat inside source list NAT_SURCHARGE_VLAN60 interface Serial0/3/0 overload
ip nat inside source list NAT_SURCHARGE_VLAN70 interface Serial0/3/1 overload

ip access-list standard NAT_SURCHARGE_VLAN60
 permit 192.168.60.0 0.0.0.255
ip access-list standard NAT_SURCHARGE_VLAN70
 permit 192.168.70.0 0.0.0.255
```

Bâtiment Rouge :

Routage :

```
Routeur-ROUGE#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 132.186.32.37 to network 0.0.0.0

     132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       132.186.32.32/30 is directly connected, Serial0/3/0
L       132.186.32.34/32 is directly connected, Serial0/3/0
C       132.186.32.36/30 is directly connected, Serial0/3/1
L       132.186.32.38/32 is directly connected, Serial0/3/1
O       132.186.32.40/30 [110/128] via 132.186.32.33, 00:54:28, Serial0/3/0
                         [110/128] via 132.186.32.37, 00:54:28, Serial0/3/1
O    192.168.10.0/24 [110/3] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
O    192.168.20.0/24 [110/3] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
O    192.168.30.0/24 [110/3] via 192.168.170.1, 00:53:53, GigabitEthernet0/0
O    192.168.40.0/24 [110/3] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
O    192.168.50.0/24 [110/3] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
O    192.168.110.0/24 [110/3] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
     192.168.150.0/30 is subnetted, 1 subnets
O       192.168.150.0/30 [110/2] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
     192.168.160.0/30 is subnetted, 1 subnets
O       192.168.160.0/30 [110/2] via 192.168.170.1, 00:54:03, GigabitEthernet0/0
     192.168.170.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.170.0/30 is directly connected, GigabitEthernet0/0
L       192.168.170.2/32 is directly connected, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 132.186.32.37
```

```
Routeur-ROUGE#show ipv6 route
IPv6 Routing Table - 19 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
O   2001:0:CAFE:10::/64 [110/3]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
O   2001:0:CAFE:20::/64 [110/3]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
O   2001:0:CAFE:30::/64 [110/3]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
O   2001:0:CAFE:40::/64 [110/3]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
O   2001:0:CAFE:50::/64 [110/3]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
O   2001:0:CAFE:60::/64 [110/65]
     via FE80::204:9AFF:FEB6:4E01, Serial0/3/0
O   2001:0:CAFE:70::/64 [110/65]
     via FE80::204:9AFF:FEB6:4E01, Serial0/3/0
O   2001:0:CAFE:80::/64 [110/65]
     via FE80::20A:F3FF:FEAE:8BD2, Serial0/3/1
O   2001:0:CAFE:110::/64 [110/3]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
O   2001:0:CAFE:120::/64 [110/65]
     via FE80::20A:F3FF:FEAE:8BD2, Serial0/3/1
O   2001:0:CAFE:150::/64 [110/2]
     via FE80::2E0:B0FF:FE1A:120A, GigabitEthernet0/0
C   2001:0:CAFE:170::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:0:CAFE:170::2/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:0:CAFE:200::/64 [0/0]
     via Serial0/3/0, directly connected
L   2001:0:CAFE:200::2/128 [0/0]
     via Serial0/3/0, receive
C   2001:0:CAFE:210::/64 [0/0]
     via Serial0/3/1, directly connected
L   2001:0:CAFE:210::2/128 [0/0]
     via Serial0/3/1, receive
O   2001:0:CAFE:220::/64 [110/128]
     via FE80::20A:F3FF:FEAE:8BD2, Serial0/3/1
     via FE80::204:9AFF:FEB6:4E01, Serial0/3/0
L   FF00::/8 [0/0]
     via Null0, receive
```

Sous-interfaces :

```
interface GigabitEthernet0/0
 description Lien Routeur ROUGE - CORE
 ip address 192.168.170.2 255.255.255.252
 ip nat inside
 duplex auto
 speed auto
 ipv6 address 2001:0:CAFE:170::2/64
 ipv6 ospf 1 area 0
```

```
interface Serial0/3/0
 description Lien Routeur-ROUGE Routeur-BLEU
 ip address 132.186.32.34 255.255.255.252
 ip nat outside
 ipv6 address 2001:0:CAFE:200::2/64
 ipv6 ospf 1 area 0
!
interface Serial0/3/1
 description Lien Routeur-ROUGE Routeur-VERT
 ip address 132.186.32.38 255.255.255.252
 ip nat outside
 ipv6 address 2001:0:CAFE:210::2/64
 ipv6 ospf 1 area 0
 clock rate 2000000
```

NAT :

```
ip nat inside source list NAT_SURCHARGE_BATIMENT_ROUGE interface Serial0/3/1 overload
ip nat inside source list NAT_SURCHARGE_BATIMENT_ROUGE(2) interface Serial0/3/0 overload
ip classless
ip route 0.0.0.0 0.0.0.0 132.186.32.37
!
ip flow-export version 9
!
!
ip access-list standard NAT_SURCHARGE_BATIMENT_ROUGE
 permit 192.168.10.0 0.0.0.255
 permit 192.168.20.0 0.0.0.255
 permit 192.168.30.0 0.0.0.255
 permit 192.168.40.0 0.0.0.255
 permit 192.168.50.0 0.0.0.255
 permit 192.168.110.0 0.0.0.255
ip access-list standard NAT_SURCHARGE_BATIMENT_ROUGE(2)
 permit 192.168.10.0 0.0.0.255
 permit 192.168.20.0 0.0.0.255
 permit 192.168.30.0 0.0.0.255
 permit 192.168.40.0 0.0.0.255
 permit 192.168.50.0 0.0.0.255
 permit 192.168.110.0 0.0.0.255
```

Bâtiment Vert :

Routage :

```
Routeur-VERT#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     132.186.0.0/16 is variably subnetted, 5 subnets, 2 masks
O       132.186.32.32/30 [110/128] via 132.186.32.38, 00:57:48, Serial0/3/0
                         [110/128] via 132.186.32.41, 00:57:48, Serial0/3/1
C       132.186.32.36/30 is directly connected, Serial0/3/0
L       132.186.32.37/32 is directly connected, Serial0/3/0
C       132.186.32.40/30 is directly connected, Serial0/3/1
L       132.186.32.42/32 is directly connected, Serial0/3/1
     192.168.80.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.80.0/24 is directly connected, GigabitEthernet0/1.80
L       192.168.80.254/32 is directly connected, GigabitEthernet0/1.80
     192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.120.0/24 is directly connected, GigabitEthernet0/0.120
L       192.168.120.254/32 is directly connected, GigabitEthernet0/0.120
```

```
Routeur-VERT#show ipv6 route
IPv6 Routing Table - 20 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
       U - Per-user Static route, M - MIPv6
       I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
       D - EIGRP, EX - EIGRP external
O   2001:0:CAFE:10::/64 [110/67]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:20::/64 [110/67]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:30::/64 [110/67]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:40::/64 [110/67]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:50::/64 [110/67]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:60::/64 [110/65]
     via FE80::204:9AFF:FEB6:4E01, Serial0/3/1
O   2001:0:CAFE:70::/64 [110/65]
     via FE80::204:9AFF:FEB6:4E01, Serial0/3/1
C   2001:0:CAFE:80::/64 [0/0]
     via GigabitEthernet0/1.80, directly connected
L   2001:0:CAFE:80::254/128 [0/0]
     via GigabitEthernet0/1.80, receive
O   2001:0:CAFE:110::/64 [110/67]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
C   2001:0:CAFE:120::/64 [0/0]
     via GigabitEthernet0/0.120, directly connected
L   2001:0:CAFE:120::254/128 [0/0]
     via GigabitEthernet0/0.120, receive
O   2001:0:CAFE:150::/64 [110/66]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:170::/64 [110/65]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
O   2001:0:CAFE:200::/64 [110/128]
     via FE80::210:11FF:FEAC:B14A, Serial0/3/0
     via FE80::204:9AFF:FEB6:4E01, Serial0/3/1
C   2001:0:CAFE:210::/64 [0/0]
     via Serial0/3/0, directly connected
L   2001:0:CAFE:210::1/128 [0/0]
     via Serial0/3/0, receive
C   2001:0:CAFE:220::/64 [0/0]
     via Serial0/3/1, directly connected
L   2001:0:CAFE:220::2/128 [0/0]
     via Serial0/3/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

Sous-interfaces :

```
interface GigabitEthernet0/0.120
 description Vlan 120 - Serveur_Web
 encapsulation dot1Q 120
 ip address 192.168.120.254 255.255.255.0
 ip nat inside
 ipv6 address 2001:0:CAFE:120::254/64
 ipv6 ospf 1 area 0
!
```

```
interface GigabitEthernet0/1.80
 description Vlan 80 - Reunion
 encapsulation dot1Q 80
 ip address 192.168.80.254 255.255.255.0
 ip nat inside
 ipv6 address 2001:0:CAFE:80::254/64
 ipv6 ospf 1 area 0
!
interface Serial0/3/0
 description Lien Routeur-VERT Routeur-ROUGE
 ip address 132.186.32.37 255.255.255.252
 ip nat outside
 ipv6 address 2001:0:CAFE:210::1/64
 ipv6 ospf 1 area 0
!
interface Serial0/3/1
 description Lien Routeur-VERT Routeur-BLEU
 ip address 132.186.32.42 255.255.255.252
 ip nat outside
 ipv6 address 2001:0:CAFE:220::2/64
 ipv6 ospf 1 area 0
'
```

NAT :

```
ip nat inside source list NAT_SURCHARGE_VLAN80 interface Serial0/3/0 overload
ip nat inside source static tcp 192.168.120.1 80 132.186.32.42 80
ip nat inside source static tcp 192.168.120.1 443 132.186.32.42 443
ip classless
!
ip flow-export version 9
!
!
ip access-list standard NAT_SURCHARGE_VLAN80
 permit 192.168.80.0 0.0.0.255
```

4. Configuration des switchs (comprenant les paramètres DHCP, les VLAN, le LACP et les ACL) :

a. Configuration des switchs de type layer 3 :

Switch Core :

```
interface GigabitEthernet1/0/1
 no switchport
 ip address 192.168.150.2 255.255.255.252
 duplex auto
 speed auto
 ipv6 address 2001:0:CAFE:150::2/64
 ipv6 enable
 ipv6 ospf 1 area 0
!
interface GigabitEthernet1/0/2
 no switchport
 ip address 192.168.160.2 255.255.255.252
 duplex auto
 speed auto
 ipv6 address 2001:0:CAFE:160::2/64
 ipv6 enable
!
interface GigabitEthernet1/0/3
!
interface GigabitEthernet1/0/4
!
interface GigabitEthernet1/0/5
!
interface GigabitEthernet1/0/6
!
interface GigabitEthernet1/0/7
!
interface GigabitEthernet1/0/8
!
interface GigabitEthernet1/0/9
!
interface GigabitEthernet1/0/10
 no switchport
 ip address 192.168.170.1 255.255.255.252
 duplex auto
 speed auto
 ipv6 address 2001:0:CAFE:170::1/64
 ipv6 enable
 ipv6 ospf 1 area 0
```

Switch Distributed 1 :

DHCP:

```
ip dhcp excluded-address 192.168.10.128 192.168.10.253
ip dhcp excluded-address 192.168.20.128 192.168.20.253
ip dhcp excluded-address 192.168.30.128 192.168.30.253
ip dhcp excluded-address 192.168.40.128 192.168.40.253
ip dhcp excluded-address 192.168.50.128 192.168.50.253
!
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.254
 dns-server 192.168.110.1
ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.254
 dns-server 192.168.110.1
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.254
 dns-server 192.168.110.1
ip dhcp pool VLAN40
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.254
 dns-server 192.168.110.1
ip dhcp pool VLAN50
 network 192.168.50.0 255.255.255.0
 default-router 192.168.50.254
 dns-server 192.168.110.1
```

SVI:

```
interface Vlan10
 mac-address 0009.7c8c.e404
 ip address 192.168.10.253 255.255.255.0
 ip access-group ACL_VLAN10 in
 ipv6 address 2001:0:CAFE:10::253/64
 ipv6 enable
 ipv6 ospf 1 area 0
 standby version 2
 standby 10 ip 192.168.10.254
 standby 10 priority 105
 standby 10 preempt
 standby 1 ipv6 autoconfig
 standby 1 priority 105
 standby 1 preempt
!
interface Vlan20
 mac-address 0009.7c8c.e405
 ip address 192.168.20.253 255.255.255.0
 ip access-group ACL_VLAN20 in
 ipv6 address 2001:0:CAFE:20::254/64
 ipv6 enable
 ipv6 ospf 1 area 0
 standby version 2
 standby 20 ip 192.168.20.254
 standby 20 priority 105
 standby 20 preempt
 standby 2 ipv6 autoconfig
 standby 2 priority 105
 standby 2 preempt
!
interface Vlan30
 mac-address 0009.7c8c.e406
 ip address 192.168.30.253 255.255.255.0
 ip access-group ACL_VLAN30 in
 ipv6 address 2001:0:CAFE:30::254/64
 ipv6 enable
 ipv6 ospf 1 area 0
 standby version 2
 standby 30 ip 192.168.30.254
 standby 30 priority 105
 standby 30 preempt
 standby 3 ipv6 autoconfig
 standby 3 priority 105
 standby 3 preempt
```

```
interface Vlan40
 mac-address 0009.7c8c.e407
 ip address 192.168.40.253 255.255.255.0
 ip access-group ACL_VLAN40 in
 ipv6 address 2001:0:CAFE:40::254/64
 ipv6 enable
 ipv6 ospf 1 area 0
 standby version 2
 standby 40 ip 192.168.40.254
 standby 40 priority 105
 standby 40 preempt
 standby 4 ipv6 autoconfig
 standby 4 priority 105
 standby 4 preempt
!
interface Vlan50
 mac-address 0009.7c8c.e408
 ip address 192.168.50.253 255.255.255.0
 ipv6 address 2001:0:CAFE:50::254/64
 ipv6 enable
 ipv6 ospf 1 area 0
 standby version 2
 standby 50 ip 192.168.50.254
 standby 50 priority 105
 standby 50 preempt
 standby 5 ipv6 autoconfig
 standby 5 priority 105
 standby 5 preempt
!
interface Vlan110
 mac-address 0009.7c8c.e409
 ip address 192.168.110.253 255.255.255.0
 ipv6 address 2001:0:CAFE:110::254/64
 ipv6 enable
 ipv6 ospf 1 area 0
 standby version 2
 standby 110 ip 192.168.110.254
 standby 110 priority 105
 standby 110 preempt
 standby 6 ipv6 autoconfig
 standby 6 priority 105
 standby 6 preempt
```

LACP:

```
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------------------

1      Po1(SU)           LACP   Fa0/20(P) Fa0/21(P) Fa0/22(P) Fa0/23(P)
Switch#
```

ACL:

```
ip access-list extended ACL_VLAN10
 permit tcp 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.10.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.10.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_VLAN20
 permit tcp 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.20.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.20.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_VLAN30
 permit tcp 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.30.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_VLAN40
 permit tcp 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.40.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.40.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
```

Switch Distributed 2 :

DHCP:

```
ip dhcp excluded-address 192.168.10.1 192.168.10.127
ip dhcp excluded-address 192.168.20.1 192.168.20.127
ip dhcp excluded-address 192.168.30.1 192.168.30.127
ip dhcp excluded-address 192.168.40.1 192.168.40.127
ip dhcp excluded-address 192.168.50.1 192.168.50.127
!
ip dhcp pool VLAN10
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.254
 dns-server 192.168.110.1
ip dhcp pool VLAN20
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.254
 dns-server 192.168.110.1
ip dhcp pool VLAN30
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.254
 dns-server 192.168.110.1
ip dhcp pool VLAN40
 network 192.168.40.0 255.255.255.0
 default-router 192.168.40.254
 dns-server 192.168.110.1
ip dhcp pool VLAN50
 network 192.168.50.0 255.255.255.0
 default-router 192.168.50.254
 dns-server 192.168.110.1
```

SVI:

```
interface Vlan10
 mac-address 0001.9766.e602
 ip address 192.168.10.252 255.255.255.0
 ip access-group ACL_VLAN10 in
 ipv6 address 2001:0:CAFE:10::252/64
 ipv6 enable
 standby version 2
 standby 10 ip 192.168.10.254
 standby 10 preempt
 standby 1 ipv6 autoconfig
 standby 1 preempt
!
interface Vlan20
 mac-address 0001.9766.e603
 ip address 192.168.20.252 255.255.255.0
 ip access-group ACL_VLAN20 in
 ipv6 address 2001:0:CAFE:20::252/64
 ipv6 enable
 standby version 2
 standby 20 ip 192.168.20.254
 standby 20 preempt
 standby 2 ipv6 autoconfig
 standby 2 preempt
!
interface Vlan30
 mac-address 0001.9766.e604
 ip address 192.168.30.252 255.255.255.0
 ip access-group ACL_VLAN30 in
 ipv6 address 2001:0:CAFE:30::252/64
 ipv6 enable
 standby version 2
 standby 30 ip 192.168.30.254
 standby 30 preempt
 standby 3 ipv6 autoconfig
 standby 3 preempt
```

```
interface Vlan40
 mac-address 0001.9766.e605
 ip address 192.168.40.252 255.255.255.0
 ip access-group ACL_VLAN40 in
 ipv6 address 2001:0:CAFE:40::252/64
 ipv6 enable
 standby version 2
 standby 40 ip 192.168.40.254
 standby 40 preempt
 standby 4 ipv6 autoconfig
 standby 4 preempt
!
interface Vlan50
 mac-address 0001.9766.e606
 ip address 192.168.50.252 255.255.255.0
 ipv6 address 2001:0:CAFE:50::252/64
 ipv6 enable
 standby version 2
 standby 50 ip 192.168.50.254
 standby 50 preempt
 standby 5 ipv6 autoconfig
 standby 5 preempt
!
interface Vlan110
 mac-address 0001.9766.e607
 ip address 192.168.110.252 255.255.255.0
 ipv6 address 2001:0:CAFE:110::252/64
 ipv6 enable
 standby version 2
 standby 110 ip 192.168.110.254
 standby 110 preempt
 standby 6 ipv6 autoconfig
 standby 6 preempt
```

LACP:

```
Flags:  D - down        P - in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------------

1      Po1(SU)           LACP   Fa0/20(P) Fa0/21(P) Fa0/22(P) Fa0/23(P)
```

ACL:

```
ip access-list extended ACL_VLAN10
 permit tcp 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.10.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.10.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_VLAN20
 permit tcp 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.20.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.20.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_VLAN30
 permit tcp 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.30.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.30.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
ip access-list extended ACL_VLAN40
 permit tcp 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255 established
 permit udp 192.168.40.0 0.0.0.255 host 192.168.110.1 eq domain
 deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
 deny ip 192.168.40.0 0.0.0.255 192.168.110.0 0.0.0.255
 permit ip any any
```

b. **Configuration des switchs de type layer 2 :**

Switch Serveur WEB :

```
interface FastEthernet0/1
 switchport access vlan 120
 switchport mode access
!
```

```
!
interface GigabitEthernet0/1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 120
 switchport mode trunk
!
```

Switch Réunion:

```
interface FastEthernet0/1
 switchport access vlan 80
 switchport mode access
!
```

```
interface GigabitEthernet0/1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 80
 switchport mode trunk
!
```

Switch Accueil:

```
!
interface FastEthernet0/1
 switchport access vlan 60
 switchport mode access
!
```

```
interface GigabitEthernet0/1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 60
 switchport mode trunk
!
```

Switch Showroom:

```
!
interface FastEthernet0/1
 switchport access vlan 70
 switchport mode access
!
```

```
interface GigabitEthernet0/1
 switchport trunk native vlan 100
 switchport trunk allowed vlan 70
 switchport mode trunk
!
```

Switch Direction:

```
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
!
```

```
!
interface GigabitEthernet0/1
 switchport access vlan 10
 switchport mode access
!
interface GigabitEthernet0/2
 switchport access vlan 10
 switchport mode access
```

Switch RH:

```
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 20
 switchport mode access
!
interface GigabitEthernet0/1
 switchport access vlan 20
 switchport mode access
!
interface GigabitEthernet0/2
 switchport access vlan 20
 switchport mode access
!
```

Switch Commerciaux/Chargés de clientèle:

```
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 30
 switchport mode access
!
interface GigabitEthernet0/1
 switchport access vlan 30
 switchport mode access
!
interface GigabitEthernet0/2
 switchport access vlan 30
 switchport mode access
```

Switch Finance :

```
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 40
 switchport mode access
!
interface GigabitEthernet0/1
 switchport access vlan 40
 switchport mode access
!
interface GigabitEthernet0/2
 switchport access vlan 40
 switchport mode access
```

Switch Service Informatique :

```
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport access vlan 50
 switchport mode access
!
interface FastEthernet0/2
 switchport access vlan 50
 switchport mode access
!
interface GigabitEthernet0/1
 switchport access vlan 50
 switchport mode access
!
interface GigabitEthernet0/2
 switchport access vlan 50
 switchport mode access
```

Switch du serveur DNS :

```
interface FastEthernet0/1
 switchport access vlan 110
 switchport mode access
!
interface GigabitEthernet0/1
 switchport access vlan 110
 switchport mode access
!
interface GigabitEthernet0/2
 switchport access vlan 110
 switchport mode access
```